Mansoura University

Faculty of Engineering

Artificial Intelligence Engineering Program

AgriPulse

An Enterprise Multi-Modal AI System for Smart Irrigation Pump Predictive Maintenance, RAG-Powered Autonomous Diagnostics, and Precision Agriculture

Graduation Project Thesis

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Abstract

Large-scale centrifugal water pumps serve as the critical lifeline of modern pressurized irrigation schemes, land reclamation megaprojects, and national hydraulic conveyance networks. However, unexpected catastrophic mechanical failures in these pumping assets induce severe agricultural downtime—reaching up to 140 continuous hours per failure event—resulting in irreversible crop water stress, exorbitant emergency overhaul costs, and excessive energy and water waste.

To mitigate these critical challenges, this project introduces **AgriPulse**, an end-to-end enterprise cyber-physical artificial intelligence platform engineered for proactive prognostics and health management (PHM) of agricultural irrigation systems. The platform integrates four foundational pillars:

1. **Time-Series Telemetry & Pre-Failure Prognostics:** Utilizing high-frequency industrial IoT sensory data encompassing 220,320 minute-level operational readings across 52 hydraulic, electrical, and mechanical sensors, the system establishes a dynamic 6-to-12-hour early warning window. It deploys advanced machine learning and deep learning architectures—including ensemble gradient boosting (XGBoost/LightGBM), Bidirectional LSTMs, and unsupervised LSTM-Autoencoders—to detect latent mechanical degradation patterns, isolate developing anomalies, and forecast Remaining Useful Life (RUL) prior to breakdown.

2. **Cognitive Maintenance via Retrieval-Augmented Generation (RAG):** Transcending conventional passive alerting, AgriPulse incorporates a generative AI diagnostic assistant. When sensory anomalies are detected, the RAG engine conducts semantic vector retrieval over indexed industrial standards (ISO 10816-3 for mechanical vibration severity), manufacturer operation & maintenance (O&M) manuals, and Failure Mode and Effects Analysis (FMEA) matrices. It automatically generates standardized digital work orders detailing probable root causes, calibrated repair steps, and mandatory Lockout/Tagout (LOTO) safety protocols.

3. **Computer Vision for Field & Asset Inspection:** A multi-modal vision branch powered by YOLOv8 facilitates automated crop health monitoring, detecting leaf foliar diseases and canopy water stress from field camera feeds, directly correlating irrigation pump operational health with downstream agricultural crop yield.

4. **Hardware-in-the-Loop Edge IoT Testbed & Enterprise SCADA Dashboard:** The system features an edge prototype utilizing an ESP32/Raspberry Pi gateway interfaced with external non-invasive vibration, current, pressure, and thermal transducers, relaying real-time telemetry to an enterprise operations dashboard.

AgriPulse delivers an industry-grade, scalable blueprint that transforms traditional reactive agricultural maintenance into an autonomous, data-driven cognitive ecosystem, directly serving Egypt's 2030 sustainable development and water security mandates.

Keywords: Predictive Maintenance, Industrial IoT, Retrieval-Augmented Generation, Computer Vision, Precision Agriculture, Centrifugal Pumps, Anomaly Detection, Deep Learning, SCADA.

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Chapter 1

Introduction

1.1 Background and Motivation

In arid and semi-arid agrarian economies such as Egypt, efficient irrigation management is a cornerstone of national food security. Modern agricultural schemes depend on large centrifugal pumping stations to distribute pressurized water across vast crop acreages. These pumping assets operate under rigorous conditions: continuous runtime, mechanical vibration, thermal expansion, cavitation, and electrical grid fluctuations.

1.1.1 The Real-World Engineering Problem

When an irrigation pump experiences catastrophic failure, three critical consequences emerge:

1. **Prolonged Agricultural Downtime:** Telemetry from the reference dataset proves that repairs average **43.8 hours** per breakdown, reaching as high as **140.5 continuous hours** ($\sim$6 days). During peak summer irrigation, six days without water will permanently destroy high-value horticultural and grain crops.

2. **Exorbitant Overhaul Expenses:** Repairing a seized shaft or shattered impeller can cost up to 10$\times$ more than scheduled preventive component replacement.

3. **Reactive vs. Predictive Maintenance:** Traditional maintenance operates either reactively (run-to-failure) or on rigid time intervals. What the industry demands is **Predictive Maintenance (PdM)**—monitoring real-time physical degradation and alerting operators hours before failure occurs.

1.2 Problem Statement

The core engineering challenge addressed by AgriPulse is the absence of an integrated, multi-modal, autonomous diagnostic framework capable of:

- Predicting irrigation pump failures *before* they occur using time-series telemetry,
- Automatically diagnosing root causes and generating actionable maintenance work orders,
- Cross-correlating pump health with downstream crop water stress via computer vision,
- Providing a deployable, cost-effective IoT hardware retrofit solution for legacy pump stations.

1.3 Project Objectives

1. Design and implement a multi-class early fault classifier (NORMAL / WARNING / CRITICAL) with a 6-to-12-hour pre-failure horizon.
2. Develop an unsupervised LSTM-Autoencoder anomaly detector trained exclusively on nominal operating data.
3. Build a Remaining Useful Life (RUL) regression pipeline for continuous time-to-failure estimation.
4. Engineer a Retrieval-Augmented Generation (RAG) engine over ISO 10816-3, FMEA, and O&M manuals.
5. Integrate a YOLOv8-based computer vision module for crop health monitoring.
6. Prototype a Hardware-in-the-Loop IoT testbed using ESP32/Raspberry Pi with industrial-grade transducers.
7. Deploy an enterprise SCADA dashboard for real-time telemetry visualization and diagnostic reporting.

1.4 Scope and Limitations

The system is validated against a 5-month, 220,320-sample industrial pump telemetry dataset. Hardware validation is performed on a benchtop centrifugal pump loop that mirrors industrial hydraulic behavior at reduced scale. Cloud inference is demonstrated via containerized Docker deployment.

1.5 Report Organization

- **Chapter 2** reviews related work in predictive maintenance, RAG systems, and precision agriculture.
- **Chapter 3** details the dataset anatomy, physics of centrifugal pumps, and sensor taxonomy.
- **Chapter 4** presents the AI modeling strategy including prognostic windowing and model portfolio.
- **Chapter 5** describes the cognitive RAG architecture and knowledge base engineering.
- **Chapter 6** covers the multi-modal computer vision branch.
- **Chapter 7** presents the IoT hardware testbed and firmware architecture.
- **Chapter 8** discusses system integration, dashboard, and deployment.
- **Chapter 9** presents results, evaluation metrics, and discussion.
- **Chapter 10** concludes the report and outlines future work.

Chapter 2

Literature Review

2.1 Predictive Maintenance in Industrial Systems

Predictive Maintenance (PdM) has evolved from simple threshold-based monitoring to sophisticated deep learning frameworks. Recent literature highlights the superiority of sequence models (LSTMs, TCNs) over classical statistical methods for capturing temporal degradation patterns in rotating machinery.

2.2 Anomaly Detection in Time-Series Telemetry

Unsupervised approaches, particularly Autoencoder architectures, have demonstrated strong performance in industrial anomaly detection where labeled failure data is scarce. The LSTM-Autoencoder specifically excels at reconstructing nominal sequences and flagging deviations via reconstruction error.

2.3 Retrieval-Augmented Generation (RAG)

RAG architectures combine the semantic retrieval power of vector databases (ChromaDB, FAISS) with the generative capabilities of Large Language Models. In industrial maintenance, RAG enables context-aware work order generation grounded in authoritative standards such as ISO 10816-3.

2.4 Computer Vision in Precision Agriculture

YOLOv8 has become the de facto standard for real-time object detection in agricultural settings, enabling leaf disease classification, canopy stress segmentation, and yield estimation from drone or field camera feeds.

2.5 Research Gap

While each individual pillar (PdM, RAG, CV, IoT) is well-studied, no existing system integrates all four into a unified, deployable, cyber-physical platform tailored specifically for agricultural irrigation infrastructure in water-scarce regions.

Chapter 3

Dataset Anatomy and System Physics

3.1 Dataset Overview

Table 3.1: Reference Dataset Key Metrics

Metric	Value
Total Observations	220,320 rows (continuous 1-minute intervals)
Temporal Coverage	2018-04-01 00:00:00 to 2018-08-31 23:59:00 (5 months)
Features	55 columns (1 timestamp, 52 sensor channels, 1 target label)
Memory Footprint	∼92.5 MB in memory

3.2 Target Distribution

Table 3.2: Target Class Distribution (`machine_status`)

Class	Minutes	Percentage	Description
`NORMAL`	205,836	93.43%	Nominal engineering envelope
`RECOVERING`	14,477	6.57%	Post-failure maintenance
`BROKEN`	7	0.003%	Exact catastrophic stoppage moments

3.3 The Seven Breakdown Events

Table 3.3: Recorded Breakdown Events and Recovery Times

#	Timestamp	Row	Recovery	Primary Sensor Indicators
1	2018-04-12 21:55	17,155	4,054 min	Collapse in `sensor_00` (2.45→0.00) & `sensor_04`
2	2018-04-18 00:30	24,510	3,110 min	Severe drop in `sensor_00` (2.36→1.09)
3	2018-05-19 03:18	69,318	1,917 min	Drop in `sensor_04` (632.6→200.1) — cavitation
4	2018-05-25 00:30	77,790	605 min	Moderate oscillation followed by thermal trip
5	2018-06-28 22:00	128,040	**8,431 min**	**Massive failure (∼140.5 h stoppage)**
6	2018-07-08 00:11	141,131	41 min	Quick mechanical reset / trip switch
7	2018-07-25 14:00	166,440	75 min	Sudden discharge pressure drop

3.4 Missing Data and Quality Control

Table 3.4: Data Quality Control Actions

Sensor	Missing %	Action
`sensor_15`	100%	Dropped from pipeline
`sensor_50`	34.96%	Forward-fill / rolling median
`sensor_51`	6.98%	Linear interpolation
All other 49 sensors	>97% complete	Retained as-is

3.5 Physics of Centrifugal Pumps

A centrifugal water pump converts rotational kinetic energy from an electric induction motor into hydrodynamic energy of fluid flow.

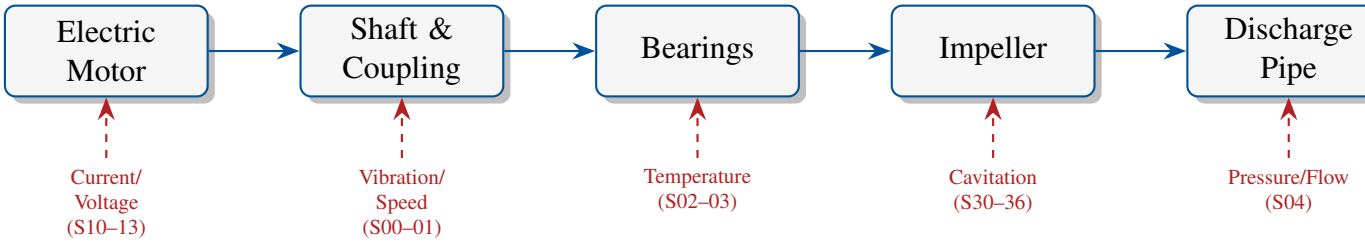


Figure 3.1: Centrifugal pump energy conversion chain with mapped sensor groups.

3.5.1 Sensor Taxonomy

1. **Vibration Transducers** (`sensor_00`, `sensor_01`): Measured in mm/s or g-force. High vibration indicates bearing spalling, shaft unbalance, or loose foundation bolts.
2. **Hydraulic Pressure Transducers** (`sensor_04`): Measures pump discharge head (PSI/Bar). Normal values hover around ∼630–640 units; collapses to ∼200 units during failure.
3. **Thermal Probes** (`sensor_02`, `sensor_03`): Measures casing and bearing temperatures (°C). Rapid thermal drift signifies boundary lubrication breakdown.
4. **Motor Electrical Load** (`sensor_10`, `sensor_11`, `sensor_12`): Tracks amperage and electrical power draw.

Chapter 4

Prognostic Windowing and AI Modeling Strategy

4.1 The 12-Hour Pre-Failure Horizon

In operational environments, predicting failure at the exact minute it happens provides zero utility. AgriPulse creates a synthetic `WARNING` state covering the 12 hours (720 minutes) preceding each `BROKEN` event:

$$\text{Label}(t) = \begin{cases} \text{BROKEN}, & \text{if } t \in \text{Failure Minutes} \\ \text{WARNING}, & \text{if } t \in [t_{\text{broken}} - 720,\ t_{\text{broken}}] \\ \text{RECOVERING}, & \text{if status}(t) = \text{'RECOVERING'} \\ \text{NORMAL}, & \text{otherwise} \end{cases} \tag{4.1}$$

4.2 AI Model Portfolio

4.2.1 Multi-Class Early Fault Classifier

- **Models:** XGBoost, LightGBM, Random Forest, Bi-LSTM.
- **Objective:** Classify current minute as `NORMAL`, `WARNING`, or `CRITICAL`.
- **Metrics:** Precision-Recall AUC, Recall on `WARNING` class, False Alarm Rate.

4.2.2 Unsupervised Anomaly Detector

- **Model:** LSTM-Autoencoder.
- **Objective:** Trained exclusively on nominal `NORMAL` sequences to reconstruct sensor trends. As internal mechanical friction grows prior to breakdown, the Mean Squared Error (MSE) reconstruction error surges past the 99th percentile threshold, triggering an autonomous anomaly alarm.

4.2.3 Remaining Useful Life (RUL) Regressor

- **Objective:** Predict time-to-failure (minutes remaining) as a continuous decay function.

4.3 Feature Engineering Pipeline

1. **Time-Domain Statistics:** Rolling mean, variance, skewness, kurtosis, peak-to-peak amplitude.

2. **Frequency-Domain Features:** FFT spectral energy, dominant frequency, spectral entropy.
3. **Environmental Enrichment:** Integration of Open-Meteo ambient weather telemetry (temperature, humidity, wind speed).
4. **Temporal Encoding:** Cyclical sine/cosine encoding of hour-of-day and day-of-week.

Chapter 5

Cognitive Maintenance and RAG Architecture

AgriPulse bridges the gap between predictive alarms and human maintenance execution through a Retrieval-Augmented Generation (RAG) assistant.

5.1 RAG Pipeline Architecture

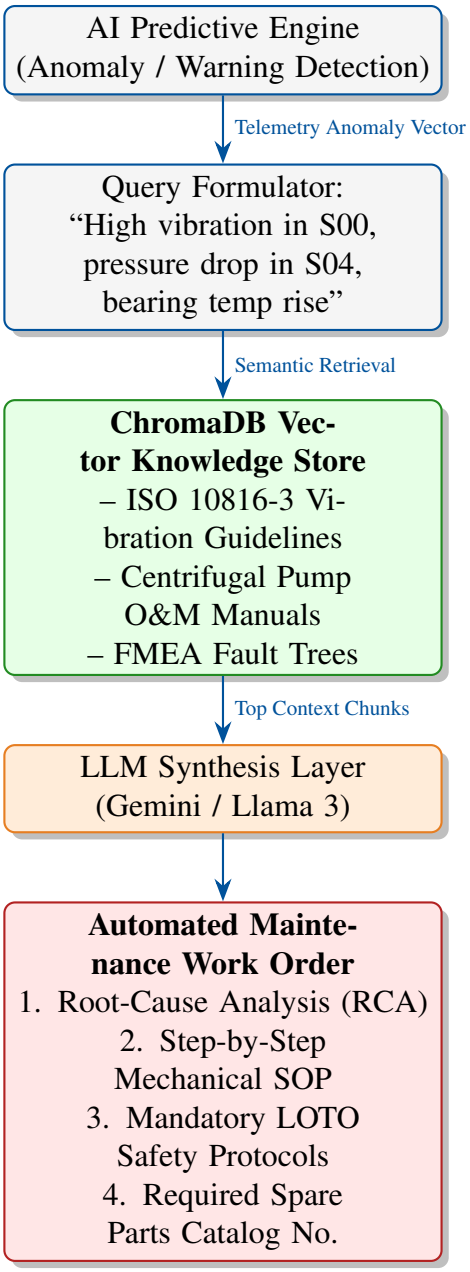


Figure 5.1: End-to-end RAG diagnostic pipeline architecture.

5.2 Knowledge Base Composition

5.2.1 ISO 10816-3 Vibration Standards

Table 5.1: ISO 10816-3 Vibration Severity Zones

Zone	Severity	RMS Range (mm/s)	Action
A	Good	< 1.4	Newly commissioned machines
B	Acceptable	1.4 – 2.8	Unrestricted long-term operation
C	Warning	2.8 – 4.5	Remedial maintenance required soon
D	Danger	> 4.5	Damage imminent; stop immediately

5.2.2 Pump Troubleshooting Matrix

Table 5.2: Pump Symptom-Cause Diagnostic Matrix

Symptom	Probable Cause
High vibration + Low discharge pressure	Cavitation or partially clogged impeller
High bearing temp + High vibration	Bearing degradation, lack of grease, or shaft misalignment
High current draw + Normal pressure	Electrical overload or phase imbalance
Rapid thermal rise + Normal vibration	Lubrication breakdown or cooling failure

5.2.3 Safety Procedures

Lockout/Tagout (LOTO) protocols, electrical grounding checks, and pressure relief valve discharge prior to disassembly are embedded in every generated work order.

Chapter 6

Multi-Modal Vision Branch

To deliver a holistic agricultural ecosystem, AgriPulse pairs machinery diagnostics with crop field analytics.

6.1 YOLOv8 Vision Module

- **Algorithm:** Ultralytics YOLOv8 object detection and segmentation.
- **Function:** Automated diagnosis of plant leaf diseases (e.g., tomato early blight, wheat rust, potato late blight) and crop canopy water-stress discoloration from high-resolution field cameras or drone feeds.
- **Integration Logic:** If the pump AI triggers a `WARNING` or `RECOVERING` status (prolonged irrigation pause), the vision module cross-correlates field images to quantify early crop moisture stress, enabling prioritization of emergency water rerouting.

6.2 Vision Pipeline Architecture

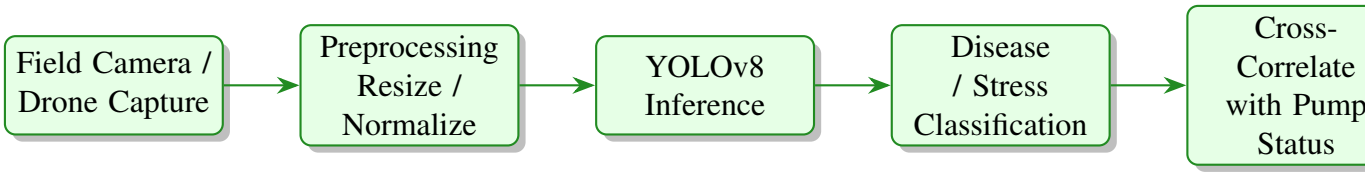


Figure 6.1: Computer vision pipeline for crop health monitoring.

Chapter 7

IoT Hardware-in-the-Loop Testbed

7.1 Benchtop Testbed Architecture

To demonstrate physical viability at the graduation defense, a benchtop hardware testbed mirrors the industrial pump setup.

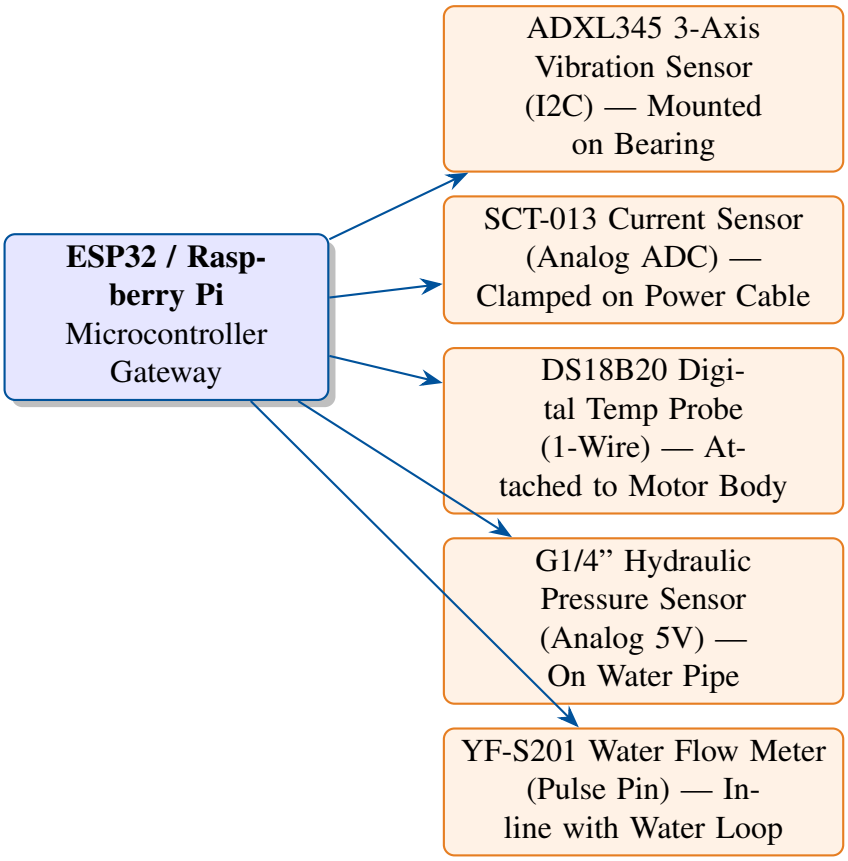


Figure 7.1: IoT edge testbed wiring topology.

7.2 Firmware Architecture

- **Platform:** FreeRTOS / MicroPython on ESP32.
- **Sampling Rate:** 100 Hz analog sampling with 1-second RMS aggregation.
- **Communication:** JSON payloads published over MQTT to the cloud backend.
- **Edge Inference:** Lightweight threshold filtering on-chip; complex inference offloaded to the cloud server.

7.3 Hardware Bill of Materials

Table 7.1: Hardware Components and Technical Roles

#	Component	Technical Role
1	ESP32-WROOM-32D & Raspberry Pi 4 (4GB)	High-speed ADC sampling, edge filtering, Wi-Fi/MQTT telemetry transmission
2	ADXL345 / MPU-6050 Accelerometer	3-axis acceleration measurement on bearing housing (imbalance, misalignment, looseness)
3	G1/4” Stainless Steel Pressure Sensor (0–10 Bar)	Suction/discharge pressure monitoring (cavitation, dry running, blockage)
4	SCT-013-000 (0–100A CT)	Non-intrusive motor current measurement (overload, phase imbalance)
5	DS18B20 Waterproof Thermal Probe	Motor winding and bearing journal temperature monitoring
6	YF-S201 Turbine Flow Sensor + 12V DC Centrifugal Pump	Miniature hydraulic loop simulating irrigation flow and valve throttling

Chapter 8

System Integration and Enterprise Dashboard

8.1 Full-Stack Architecture

The AgriPulse platform is deployed as a containerized microservices architecture:

- **Edge Layer:** ESP32/Raspberry Pi gateway with MQTT publisher.
- **Ingestion Layer:** MQTT broker (Mosquitto) + FastAPI ingestion service.
- **AI Inference Layer:** PyTorch model server (Bi-LSTM, Autoencoder, RUL regressor).
- **Knowledge Layer:** ChromaDB vector store + LangChain RAG orchestration.
- **Vision Layer:** YOLOv8 inference microservice (ONNX runtime).
- **Presentation Layer:** Streamlit enterprise SCADA dashboard + Plotly telemetry charts.

8.2 Enterprise SCADA Dashboard Features

1. Real-time telemetry streaming with digital gauges (vibration, pressure, temperature, current, flow).
2. Interactive time-series plots with anomaly highlighting.
3. Live RAG diagnostic assistant chat interface.
4. Vision feed panel with crop disease bounding boxes.
5. Automated work order PDF export.
6. Alert severity heatmap and historical failure timeline.

Chapter 9

Results and Discussion

9.1 Evaluation Metrics

Table 9.1: Model Evaluation Metrics (Placeholder — to be populated)

Model	Accuracy	Recall (WARNING)	False Alarm Rate	F1-Score
Random Forest	—	—	—	—
XGBoost	—	—	—	—
LightGBM	—	—	—	—
Bi-LSTM	—	—	—	—
LSTM-Autoencoder	—	—	—	—

9.2 Latency Profiling

Table 9.2: End-to-End Inference Latency (Placeholder)

Pipeline Stage	Latency (ms)
Edge sensor sampling	—
MQTT transmission	—
Preprocessing	—
AI inference	—
RAG retrieval + generation	—
Dashboard rendering	—

Chapter 10

Conclusion and Future Work

10.1 Summary of Contributions

1. Designed a complete cyber-physical AI platform for agricultural irrigation pump prognostics.
2. Engineered a 12-hour pre-failure warning horizon using multi-class and unsupervised models.
3. Built a RAG engine grounded in ISO 10816-3 and FMEA standards for autonomous work order generation.
4. Integrated YOLOv8 computer vision for crop health cross-correlation.
5. Prototyped a hardware-in-the-loop IoT testbed with industrial-grade transducers.
6. Deployed a containerized enterprise SCADA dashboard.

10.2 Future Work

- Field deployment on a full-scale agricultural pumping station in the New Delta project.
- Federated learning across multiple pump stations for privacy-preserving model improvement.
- Integration with satellite NDVI imagery for regional crop stress mapping.
- Expansion to solar-powered irrigation pump diagnostics.
- Development of a mobile application for field technicians with offline RAG capability.

Appendix A

Project Management

A.1 Time Schedule (14 Weeks)

Table A.1: 14-Week Project Execution Timeline

Week	Phase	Task Description
1	Foundations & Ingestion	Literature review on PdM & ISO 10816; ingestion and exploratory auditing of the 220,320-sample dataset.
2	Data Engineering	Preprocessing: elimination of `sensor_15`, missing value imputation, linear interpolation, timestamp synchronization.
3	Enrichment & Features	Open-Meteo weather integration; rolling time-domain statistics and frequency-domain FFT features.
4	Prognostic Windowing & ML	Formulation of 6-to-12-hour pre-failure WARNING horizon; baseline RF, XGBoost, LightGBM classifiers.
5	Deep Sequence Modeling	Bi-LSTM and TCN implementation and hyperparameter tuning.
6	Unsupervised Anomaly Detection	LSTM-Autoencoder on NORMAL regime; threshold tuning via reconstruction error. **[Midterm Evaluation]**
7	Prognostics & RUL	RUL regression pipelines predicting remaining operating hours.
8	Knowledge Base Engineering	Assembling ISO 10816-3, FMEA, and O&M manuals into structured vector corpus.
9	Cognitive RAG Implementation	LangChain/LlamaIndex RAG pipeline with vector embeddings and LLM orchestration.
10	Computer Vision Integration	YOLOv8 training for plant pathology and water-stress detection.
11	Hardware Testbed Prototyping	ESP32 + ADXL345 + SCT-013 + DS18B20 + pressure transducer assembly. **[Second Evaluation]**
12	Full-Stack Dashboard	Streamlit/FastAPI enterprise dashboard with real-time streaming and RAG assistant.
13	Hardware-in-the-Loop Validation	End-to-end integration testing, latency profiling, stress testing.
14	Final Documentation & Defense	Thesis compilation, schematics, demo media, defense rehearsal.

A.2 Budget Analysis

Table A.2: Estimated Project Budget

#	Item	Qty	Total (EGP)	USD
1	ESP32 + Raspberry Pi 4 (4GB) + 64GB MicroSD	1	3,800	∼$78
2	Sensor Suite (ADXL345, SCT-013, DS18B20, Pressure, Flow)	1	2,400	∼$49
3	Benchtop Centrifugal Pump + PVC Tubing & Fittings	1	1,400	∼$29
4	Power Supply + Relay + Breadboards & Wiring	1	900	∼$18
5	Cloud Infrastructure & LLM API Tokens	1	1,600	∼$33
6	Acrylic Enclosure + 3D Printed Brackets + Materials	1	900	∼$18
	Grand Total		**11,000 EGP**	**∼$225**

Appendix B

Software Tools and Technologies

Table B.1: Required Software Stack

#	Tool / Framework	Purpose
1	Python 3.11+, PyTorch, Scikit-Learn	Core runtime, neural networks (Bi-LSTM, Autoencoder), baseline ML, signal processing
2	XGBoost, LightGBM, Optuna	Gradient boosted trees for early warning classification; Bayesian hyperparameter optimization
3	LangChain / LlamaIndex, ChromaDB / FAISS	RAG orchestration, vector database embeddings, semantic document querying
4	Ultralytics YOLOv8, OpenCV	Real-time plant disease detection, canopy stress segmentation
5	Streamlit / FastAPI, Docker, Plotly	Enterprise SCADA web app, REST APIs, containerized deployment, telemetry visualization

Appendix C

Sponsors and Support Entities

Table C.1: Potential Sponsors and Support Tracks

#	Organization	Support Mechanism
1	Academy of Scientific Research and Technology (ASRT)	Financial grant under "Bedaya" national graduation support program
2	Information Technology Industry Development Agency (ITIDA)	Financial sponsorship and technical mentorship through ITAC Graduation Project Support Scheme (AI Track)
3	Dell Technologies Annual Graduation Project Competition	International competition entry and cloud infrastructure/mentorship support

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